

COOPER LIGHTING · SIGNIFY

A live mesh of devices, made *legible*.

CORE Insights turns raw lighting & sensor telemetry into operational intelligence, one platform serving the portfolio analyst, the facilities manager, and the technician on the floor.

ROLE

Lead Product Designer

PLATFORM

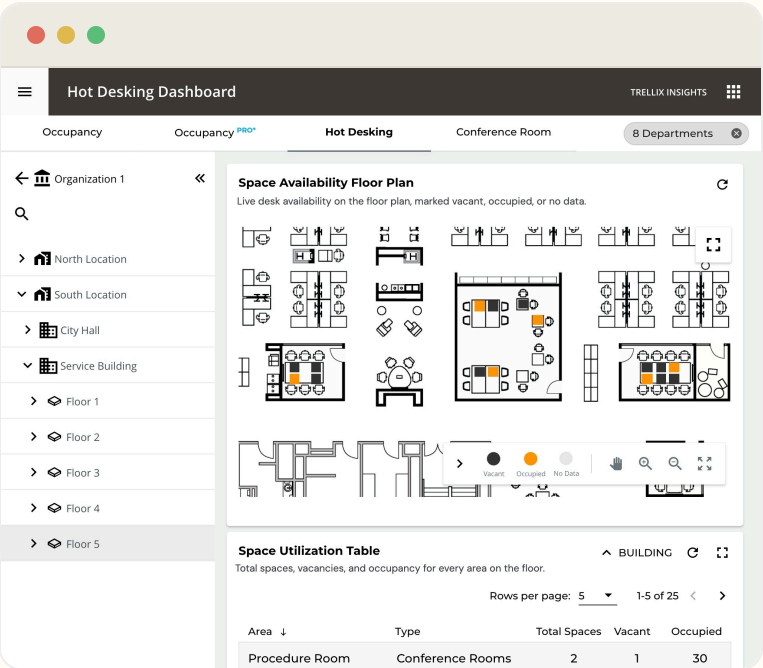
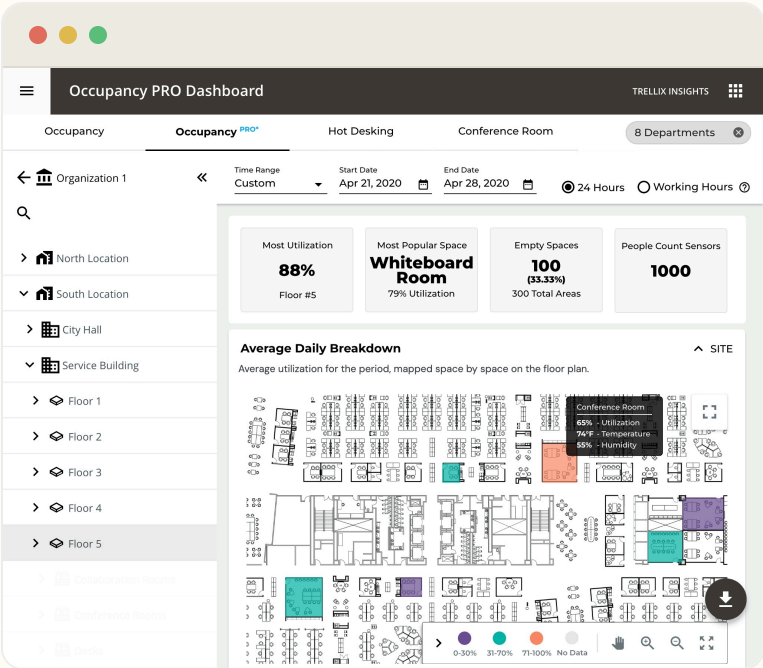
Web · iOS · Android

OUTCOME

↓23% energy · 90% faster reports

CREATED IN

Adobe XD · 2020–2021



Turning lighting data into space intelligence.

CORE Insights is a space-utilization analytics application built on the WaveLinx CORE connected-lighting platform. It transforms data from occupancy sensors embedded in commercial LED luminaires into intelligence about how building spaces are actually being used.

As Experience Design Lead I owned the end-to-end UX from initial concept — the dashboard, the reporting suite, the floor-map visualization, and the on-premises to cloud migration path. This case study focuses on the navigation and data-comprehension challenge that was keeping facilities managers from extracting value from the platform.

5

Hierarchy levels
(Client → Space)

21+

Occupancy sensors
per floor (avg)

∞

Unlimited spaces
per license

Platform	WaveLinx CORE, on-premises Smart Spaces platform with cloud extension (WaveLinx Cloud)
Application	CORE Insights, occupancy analytics dashboard and reporting suite
Users	Facilities managers, real estate planners, building operations teams, executive stakeholders
Deployment	On-premises (CORE Pro gateway) and Cloud
Data source	Occupancy sensors in WaveLinx luminaires and ceiling sensors, coordinated by WaveLinx Area Controllers (WAC)
Licensing	One-time license purchase; unlimited spaces per installation
Disciplines	UX Research, Information Architecture, Interaction Design, Data Visualization, Usability Testing

Turning an installed base into a data platform.

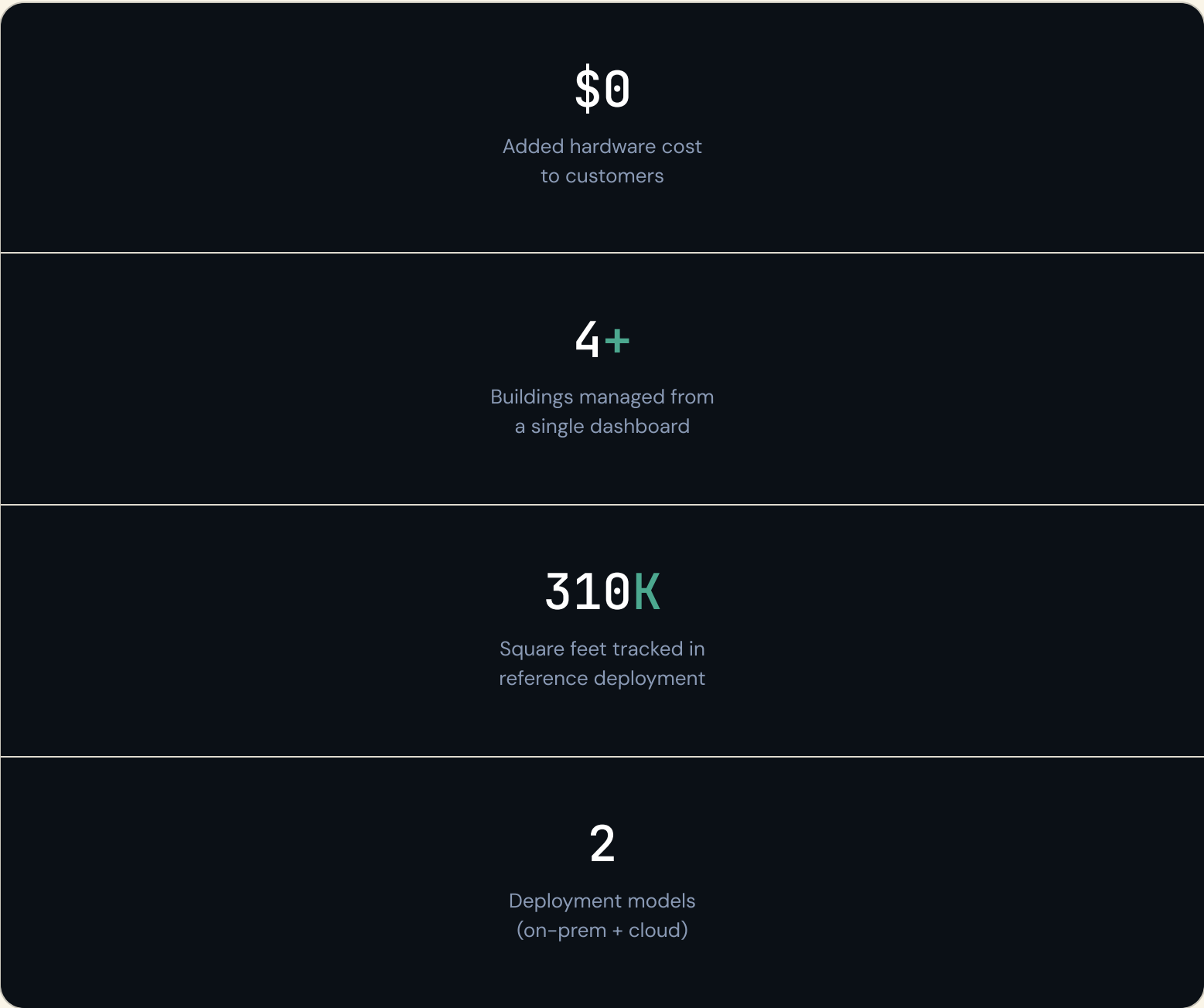
Commercial real estate was undergoing a fundamental shift in how space was valued. Post-pandemic hybrid work made historical occupancy assumptions unreliable, and facilities teams were being asked to justify space allocation with data they didn't have. Cooper Lighting was sitting on a unique asset: millions of occupancy sensors already installed in commercial buildings as part of code-compliant lighting controls. The data was being collected, it just wasn't surfaced in a way that was useful to the people making space decisions.

CORE Insights was Cooper's answer. The business case was simple: turn the lighting network into a space-intelligence platform, creating a new revenue stream from an installed base that

was already generating the data. Every WaveLinx installation became a potential Insights upsell, no additional hardware required.

THE COMPETITIVE ADVANTAGE

Dedicated workspace-analytics vendors like VergeSense, Density, and SpacelQ were gaining traction selling standalone sensor deployments. Cooper's sensors were already in the ceiling. The challenge was making the analytics layer as compelling as the purpose-built competitors, with an interface facilities managers would actually use day-to-day.



Sensors streamed occupancy, energy, and environmental data around the clock. But for the facilities teams who ran these buildings, the signal was buried in monthly PDFs and spreadsheets.

My job: design the operator's window into that live mesh, and give three very different users exactly the view they needed, without drowning any of them.



Every building was a silo

No way to compare occupancy, energy, or alerts across a portfolio. Each site lived in its own report.



Three users, one interface

PRO analysts wanted portfolio analytics. Site managers wanted floor control. Technicians wanted answers on the floor.



No shared taxonomy

Fixtures, zones, and spaces had no semantic model, so nothing could roll up or drill down cleanly.

the 1990s, the number of people in the world who are under 15 years of age is expected to increase from 1.1 billion to 1.5 billion.

As the world's population grows, the demand for food and other resources will increase. This will put pressure on the environment and on the world's food supply.

One way to meet this demand is to increase the amount of food that is produced. This can be done by using more land for agriculture, by using more water, or by using more fertilizers.

Another way to meet this demand is to reduce the amount of food that is wasted. This can be done by improving the way that food is stored and distributed, or by changing the way that people eat.

There are many other ways to meet this demand, and it is important that we find ways to do so that do not harm the environment or the world's food supply.

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Occupancy

Occupancy **PRO**

Hot Desking

Conference Room

8 Departments

Organization 1

Time Range
Real Time



> North Location

▼ South Location

> City Hall

▼ Service Building

> Floor 1

> Floor 2

> Floor 3

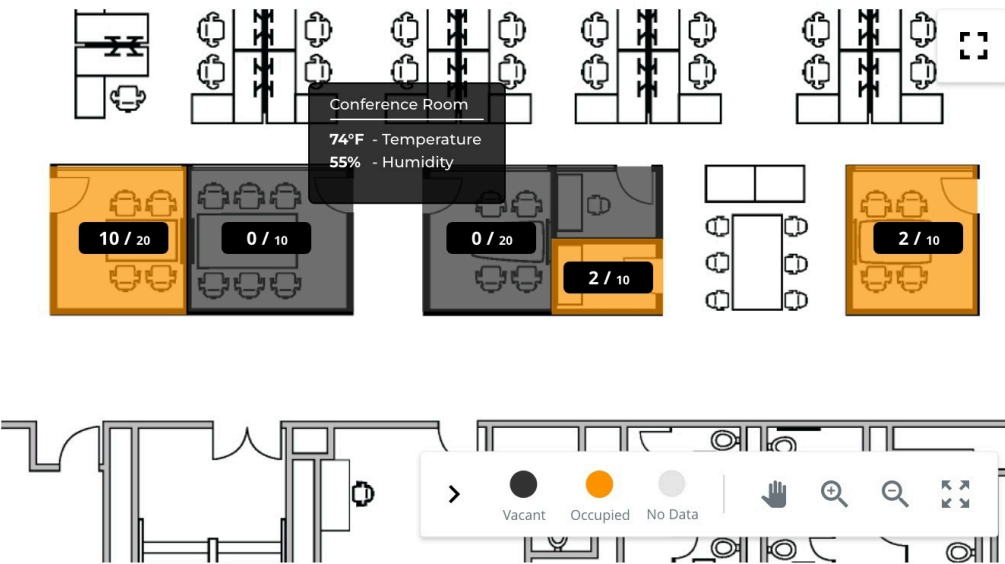
> Floor 4

> Floor 5

Utilization Floor Plan



Each space on the floor plan shows live people against capacity, with temperature and humidity on hover.



Space Utilization



Vacant, occupied, total capacity, and people right now, for every space type.

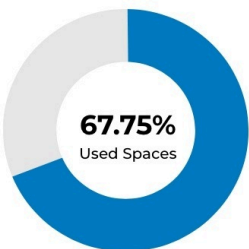
Type ↓	Vacant	Occupied	Total Capacity	Total People
Conference Rooms	2	1	30	10
Collaboration Rooms	3	2	45	8
Desks	5	4	5	4
Offices	5	4	100	47
Theatres	1	0	5	0

Space Utilization



Share of spaces in use vs sitting empty.

32.25%
Empty Spaces
31%
Last Week
16
Total Spaces

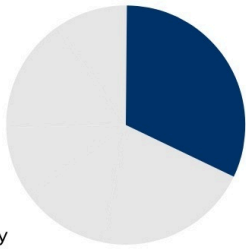


Capacity Utilization



People on site as a share of total capacity.

38.29%
Utilization
49.6%
Last Week
69 **185**
People Total Capacity



the 1990s, the number of people in the world who are under 15 years of age is expected to increase from 1.2 billion to 1.5 billion.

As the world's population grows, the demand for food and other resources will increase. This will put pressure on the environment and on the world's food supply.

One way to meet this demand is to increase the amount of food that is produced. This can be done by using more land for agriculture, or by increasing the productivity of the land that is already being used.

Another way to meet this demand is to reduce the amount of food that is wasted. This can be done by improving the way that food is stored and distributed, or by changing the way that people eat.

There are many other ways to meet the world's growing demand for food and other resources. It is up to us to decide which way is the best.

One of the most important things we can do is to make sure that we are using resources in a sustainable way. This means that we are using resources in a way that will not harm the environment or the future generations.

There are many things we can do to make sure that we are using resources in a sustainable way. We can reduce the amount of waste that we produce, we can use less energy, and we can use more recycled materials.

By making these changes, we can help to make sure that the world's food and other resources will be available for everyone in the future.

It is our responsibility to make sure that we are using resources in a sustainable way. We must all do our part to make sure that the world is a better place for everyone.

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8 Departments



Organization 1



> North Location

✓ South Location

> City Hall

✓ Service Building

> Floor 1

> Floor 2

> Floor 3

> Floor 4

✓ Floor 5

✓ Collaboration Rooms

Whiteboard Room 1

Whiteboard Room 2

> Conference Rooms

> Desks

> Offices

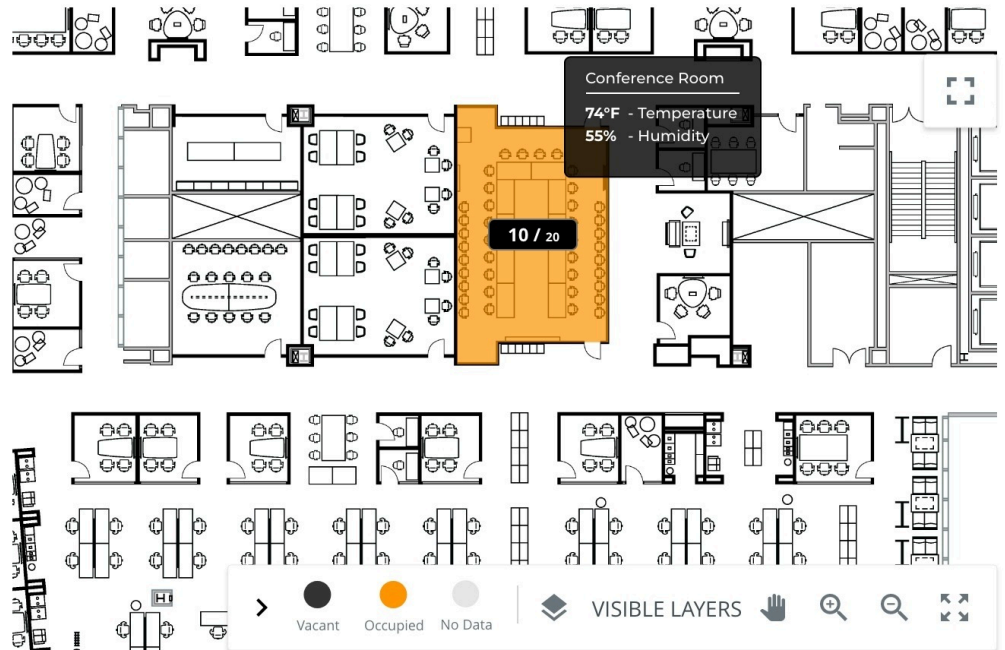
> Theatres

Occupied

Space Availability Floor Plan

^ AREA

Room status on the floor plan, with live occupancy, temperature, and humidity for the selected room.

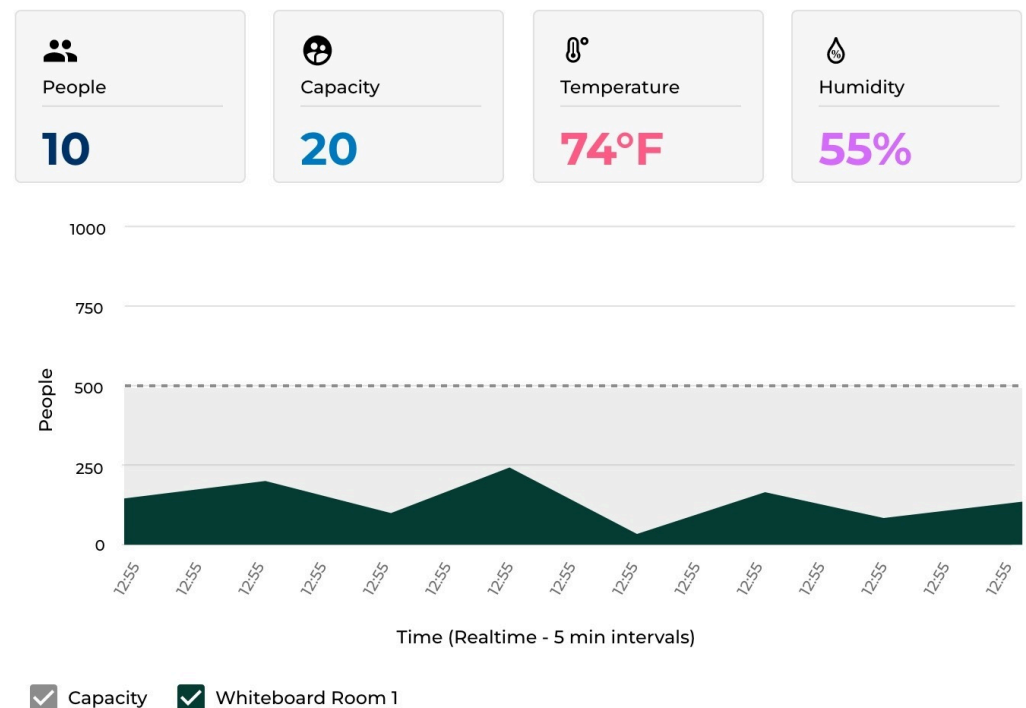


Space Availability Area Graph

^ AREA TABLE

Live people count in the room, tracked against capacity in five-minute intervals.

🕒 13:00



04 • RESEARCH APPROACH

The platform modeled every building as a strict device hierarchy. Facilities teams didn't think in device trees, they thought in questions. I ran interviews across five stakeholder groups to map the full spectrum of data-consumption needs, then measured the cost of the most common tasks.

Stakeholder interviews

STAKEHOLDER GROUP	KEY INSIGHT
Facilities Managers	Need quick answers to operational questions ("Which conference rooms on Floor 3 are underused?"); rarely drill past floor level.
Real Estate Planners	Need portfolio-level trending for lease and space planning; consume data in bulk exports and executive reports.
Building Ops / IT	Need sensor-health confirmation and data-confidence indicators; the primary concern is system reliability.
C-Suite / Executives	Need headline metrics and cost-per-square-foot justification; won't use the tool directly, they consume PDF exports.
Lighting Installers	Need to verify sensor placement and coverage during commissioning; Insights is a validation tool, not a daily-use one.

Task analysis

I mapped the five most common tasks and measured clicks, context switches, and navigation resets.

TASK	CLICKS	PAIN LEVEL
Check average occupancy for a specific floor	4-6 	Moderate
Compare occupancy across two floors	10-14 	High
Find underutilized conference rooms	7-9 	High
Generate a PDF report for a building	5-7 	Moderate
Filter by department across floors	8-12 	High

RANGE BARS SHARE ONE 0-14 CLICK SCALE

05 • MENTAL MODEL

The platform modeled every building as a strict device hierarchy. Facilities teams didn't think in device trees, they thought in *places*. I mapped that system model onto a spatial one, so each level reveals just enough, and the visualization itself changes to match the question being asked.

LEVEL 01

Organization

Compare every site at a glance.

LEVEL 02

Location

Rank campuses within a region.

BAR · COMPARISON

BAR · COMPARISON

LEVEL 03

Building

Break a site into its floors.



STACK · ROLLUP

LEVEL 04

Floor

Orient in physical space.



HEAT MAP · SPATIAL

LEVEL 05

Room

Track one space over time.



LINE · TRENDS

High level → bar charts for comparison

Floor level → heat map for spatial orientation

Room level → line charts for trends over time

THE MODEL, IN THE PRODUCT





Occupancy

Occupancy **PRO**

Hot Desking

Conference Room

8 Departments



Organization 1



Time Range

Custom

Start Date

Apr 21, 2020



End Date

Apr 28, 2020

☒ 24 Hours☐ Working Hours ?

> North Location

v South Location

> City Hall

v Service Building

> Floor 1

> Floor 2

> Floor 3

> Floor 4

> Floor 5

> Collaboration Rooms

> Conference Rooms

> Desks

> Offices

> Theatres

Most Utilization

88%

Floor #5

Most Popular Space

Whiteboard Room

79% Utilization

Empty Spaces

100
(33.33%)

300 Total Areas

People Count Sensors

1000

Average Daily Breakdown

^ SITE

Average utilization for the period, mapped space by space on the floor plan.



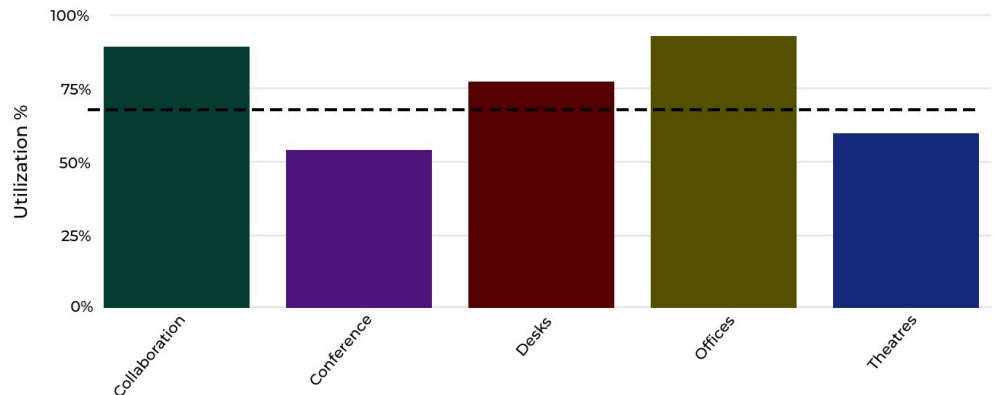
Average Daily Breakdown

^ BUILDING SORT TABLE

Average utilization by space type, measured against the building average.

OVERVIEW

TRENDING



SHOW ALL HIDE ALL

---- Average



Capacity



Collaboration



Conference



Desks



Offices



Theatres

Space Utilization

Share of spaces in use vs sitting empty.

32.25%

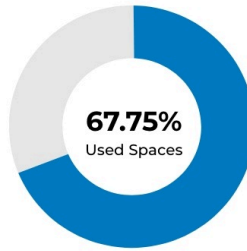
Empty Spaces

31%

Last Week

16

Total Spaces



Capacity Utilization

People on site as a share of total capacity.

38.29%

Utilization

49.6%

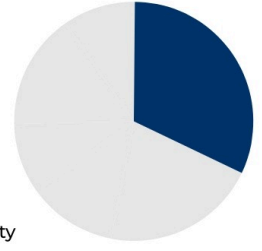
Last Week

69

People

185

Total Capacity



Total Time Space

Space over capacity

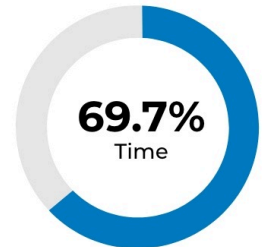
230

Hours



69.7%

Time



Refreshed on April 28, 2020 9:58am

HIGH LEVEL • BAR CHARTS

The Occupancy PRO overview, KPIs up top, utilization bars comparing space types at a glance.





Occupancy

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8 Departments



Organization 1



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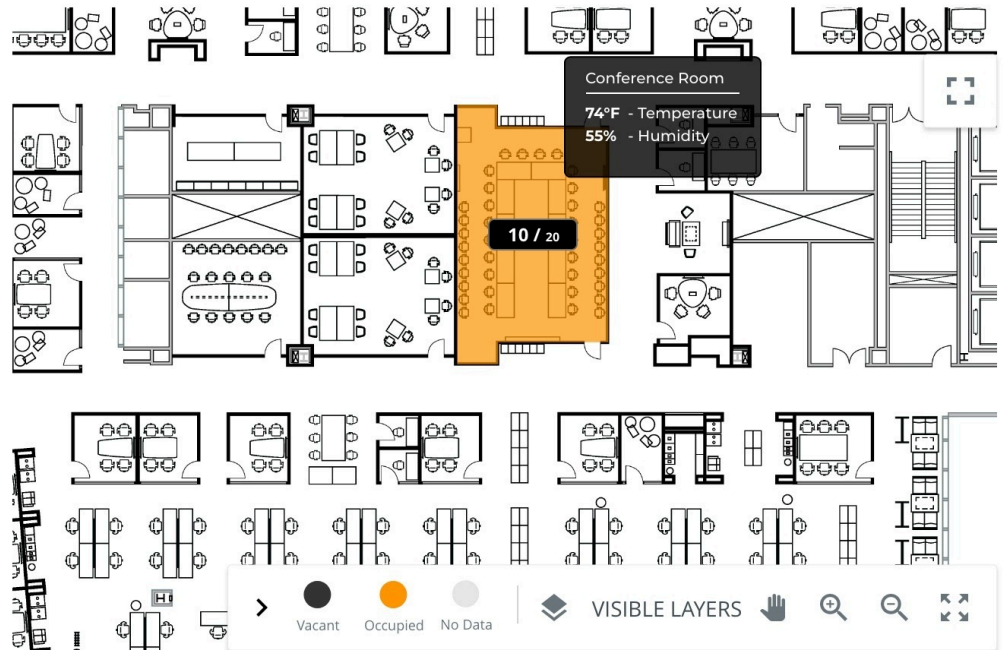
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Space Availability Floor Plan

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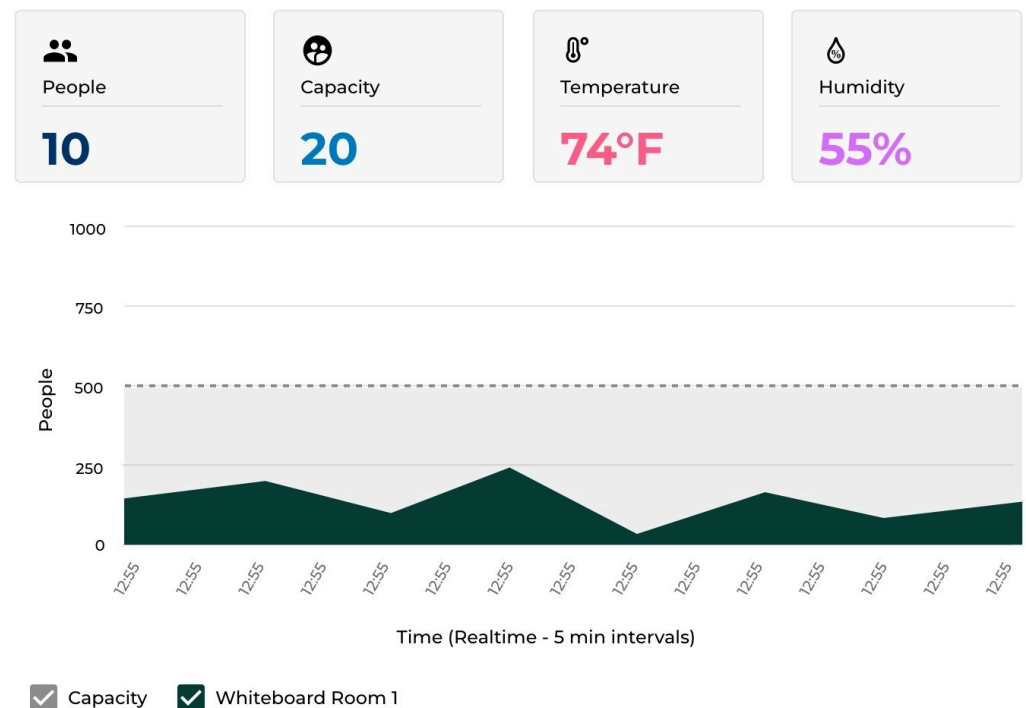


Space Availability Area Graph

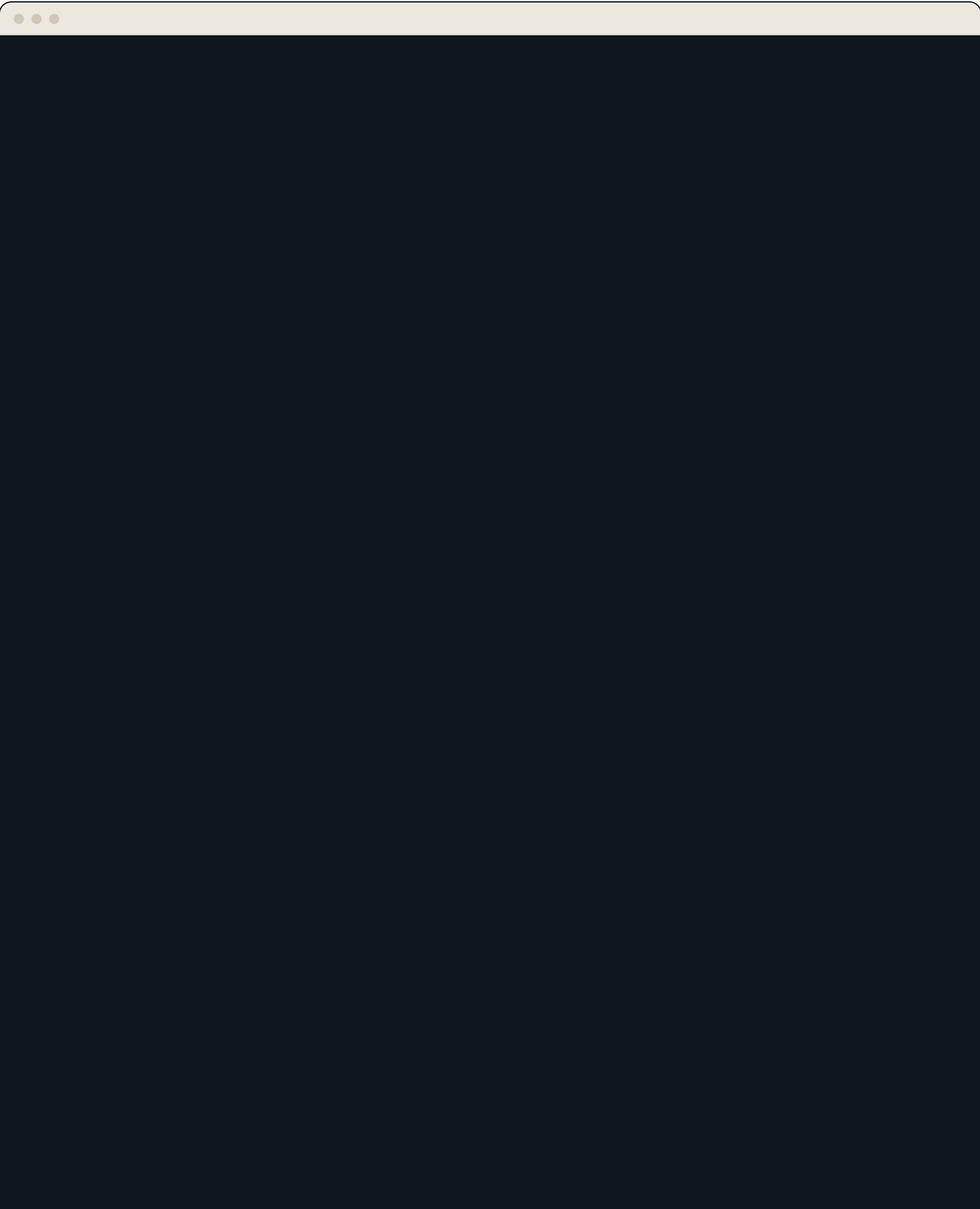
^ AREA TABLE

Live people count in the room, tracked against capacity in five-minute intervals.

🕒 13:00



One room over time, occupied status on the plan, live people, capacity and comfort, and the real-time graph.



A screenshot of the application's legend and navigation controls. The legend is a horizontal bar with three colored circles: a black circle labeled 'Vacant', an orange circle labeled 'Occupied', and a grey circle labeled 'No Data'. To the right of the legend are navigation icons: a right arrow, a hand icon, a magnifying glass icon, a zoom out icon, and a full screen icon.

Rows per page: 5 1-5 of 25 < >

Area ↓	Type	Total Spaces	Vacant	Occupied
Procedure Room	Conference Rooms	2	1	30
Exam Room 1	Desks	3	2	45
Exam Room 2	Desks	5	4	5
Exam Room 3	Desks	5	4	100
Lab	Conference Rooms	1	0	5

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
08:00							
09:00							
10:00							
11:00							
12:00							
13:00							
14:00							
15:00							
16:00							

Every desk's state on the plan, what's free right now, and the busiest-time heat map by hour.

0-20% 20%-40% 40%-60% 60%-80% 80%-100%

Refreshed on April 28, 2020 9:58am

06 · THE SOLUTION, LIVE CONSOLE

A spatial model replaced the spreadsheet. **Click any space** to inspect live occupancy, status, and trend.

Tower A · Floor 5

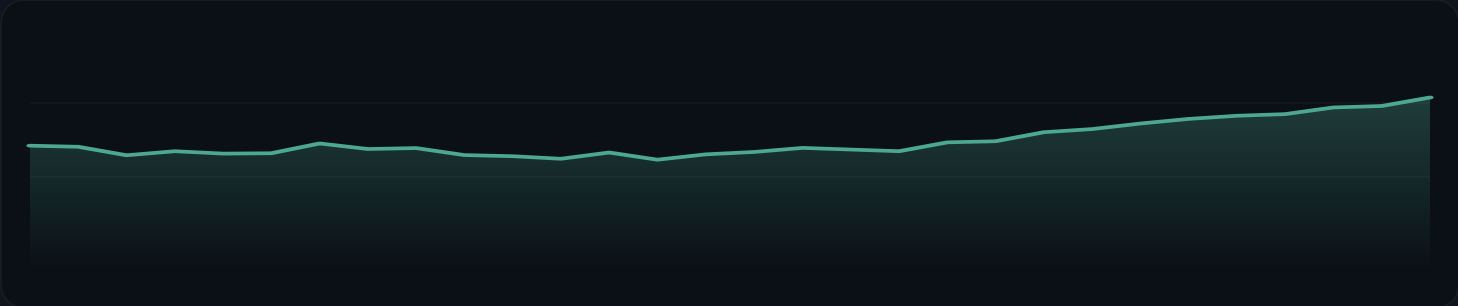
12 spaces · 342 sensors online

LIVE



Vacant Light Moderate Busy At capacity

76% ▲ peak 76%



Avg occupancy	56%
Active alerts	2

Select a space on the map to drill in →

07 • DESIGN DECISIONS

01

Progressive disclosure

Org → Location → Building → Floor → Room. Each level reveals just enough detail, keeping cognitive load manageable at the top.

02

Visualization adapts to context

Bar charts to compare, a heat map to orient in space, line charts to read trends. The chart matches the question.

03

Environment, in context

Temperature and humidity layered onto the floor plan on hover, answering "what's happening in that

04

Real-time vs. historical

Historical asks "how much was it used?" Real-time asks the simpler "is this room free *right now*?" Each

room right now?" without leaving the view.

view is shaped to its question.

05

Spatial-first at the floor

At floor level a heat map replaces the bar chart, spatial context becomes more valuable than comparative data.

06

One-click answers

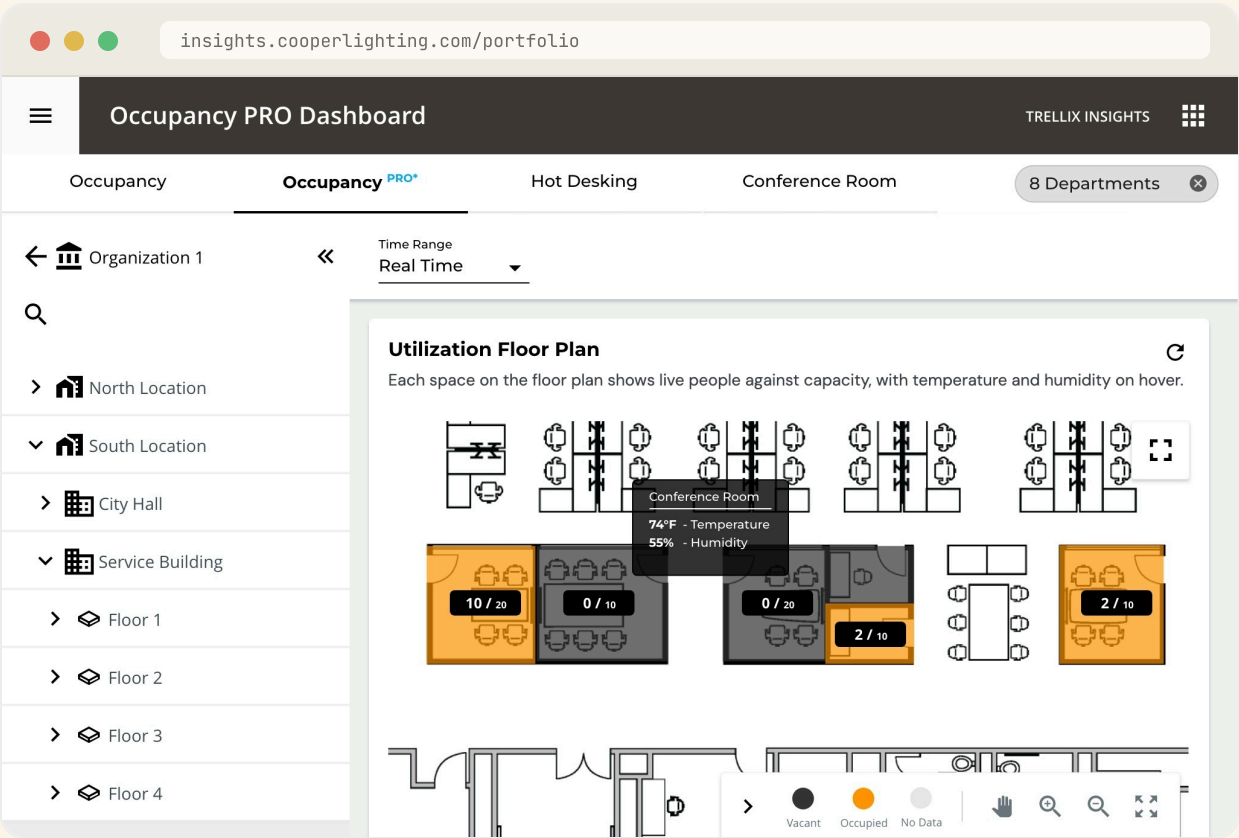
KPI cards surface "which spaces are empty" and "most popular space" up front, answers that used to take seven clicks now take one.

● ● ● ● ● ● ● → ● 7 CLICKS → 1

08 · ONE PLATFORM, ROLE BY ROLE

PRO · Portfolio

Site · Floor



PRO users manage dozens of buildings. The dashboard leads with cross-site rollups, utilization, capacity, and busiest-time heatmaps, so an analyst spots the outlier site in seconds.

- ✓ Floor-plan availability with live vacant / occupied state
- ✓ Sortable utilization tables that drill from org → building → floor
- ✓ Weekly heatmaps to plan space and staffing

09 • INFORMATION ARCHITECTURE

Rather than force one layout onto every scale, each level of the hierarchy surfaces the indicators, visualizations, and reports that fit the decision being made there, from a portfolio bar chart down to a single room's trend line.

LEVEL	KEY INDICATORS	VISUALIZATIONS	REPORTS
Client	Avg Occupancy, Empty Spaces, Buildings / Floors count, Sensors	Client Occupancy bar chart (per building)	Space Occupancy (Overview + Trending)
Building	Avg Occupancy, Empty Spaces, Floors count, Sensors	Building Occupancy bar chart (per floor)	Space Occupancy + Space Summary
Floor	Avg Occupancy, Empty Spaces, Most Occupied, Sensors	Occupancy Floor Map + Space Occupancy chart	Occupancy + Summary + Popularity (Most / Least)
Area Category	Avg Occupancy, Sensors count	Filtered Floor Map + Occupancy chart	Space Occupancy for the category

HOT DESKING • TRY IT • ONE FLOOR, TWO USERS

FACILITIES

10:00

11 of 36 desks free at 10:00

 Find me a desk

Tap any teal desk to book it — or let the app point you to the emptiest wing.

10:00

● Free ● Taken ● Yours



Hot Desking

One screen, two users at once: an employee finding a free desk *right now*, and a facilities manager reading utilization patterns to optimize the floor.

- Furniture-level seats on the real floor plan, occupied vs. free at a glance
- Spatial clusters help people navigate, "free desks top-right"
- A busiest-time heat map for planning when to come in



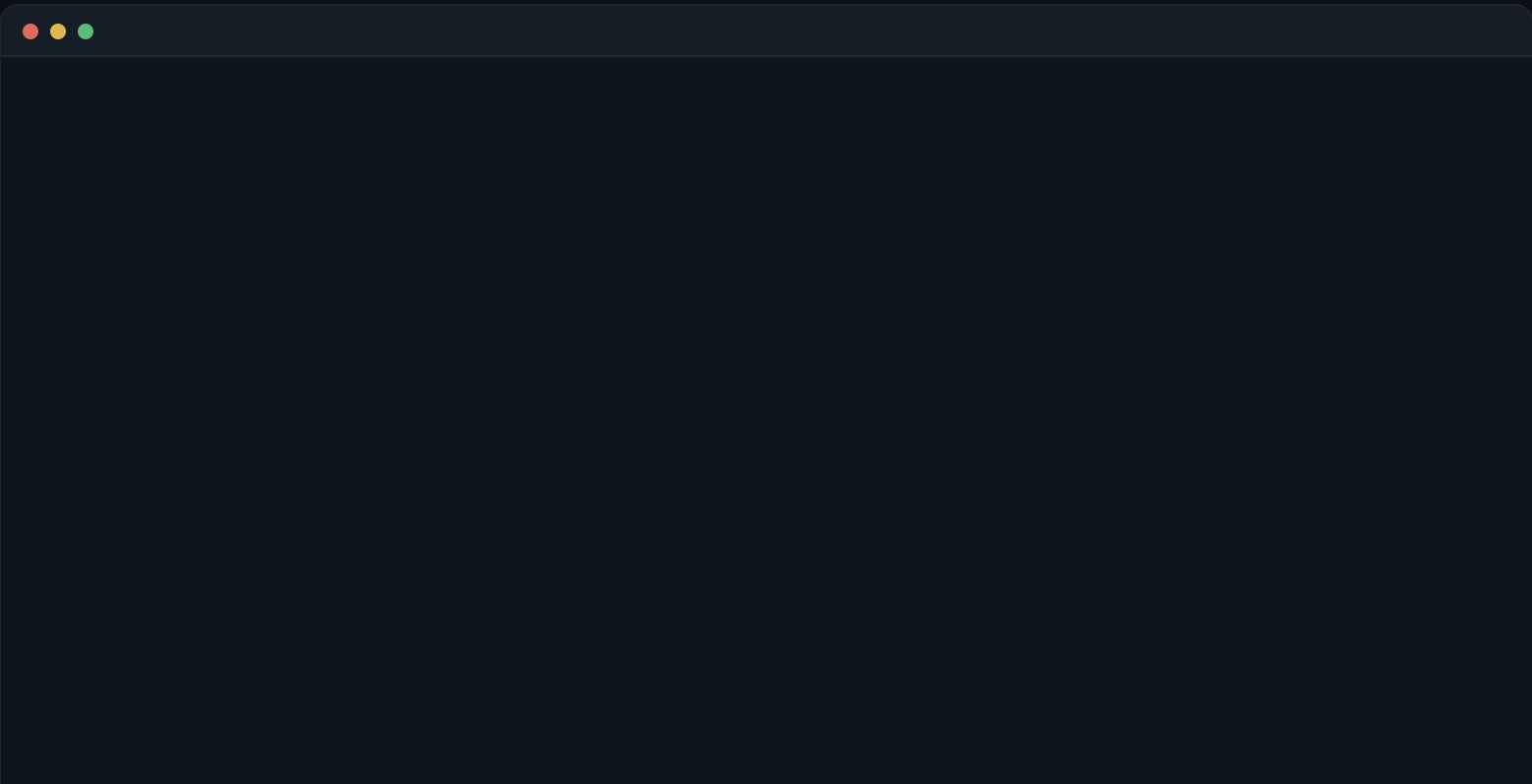
Conference Rooms

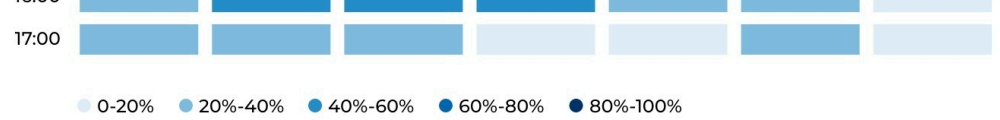
Managers watch utilization live, review it over time, and manage future bookings, with the most- and least-used rooms ranked so under-booked space is obvious.

SPACE POPULARITY · TOP 5

1	Conference Room 1		99%
2	Conference Room 2		95%
3	Conference Room 3		87%
4	Conference Room 4		68%
5	Conference Room 5		67%

SHIPPED TO THE FIELD, THE REAL SCREENS





Refreshed on April 28, 2020 9:58am

HOT DESKING

Per-desk vacancy on the floor plan, with a busiest-time heatmap for planning arrivals.





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Organization 1



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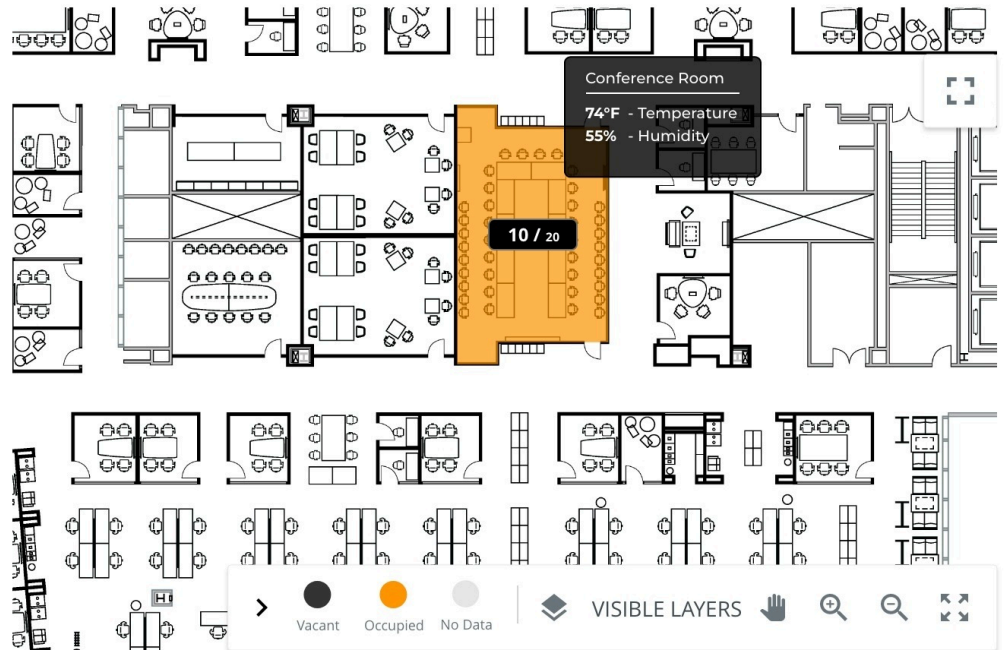
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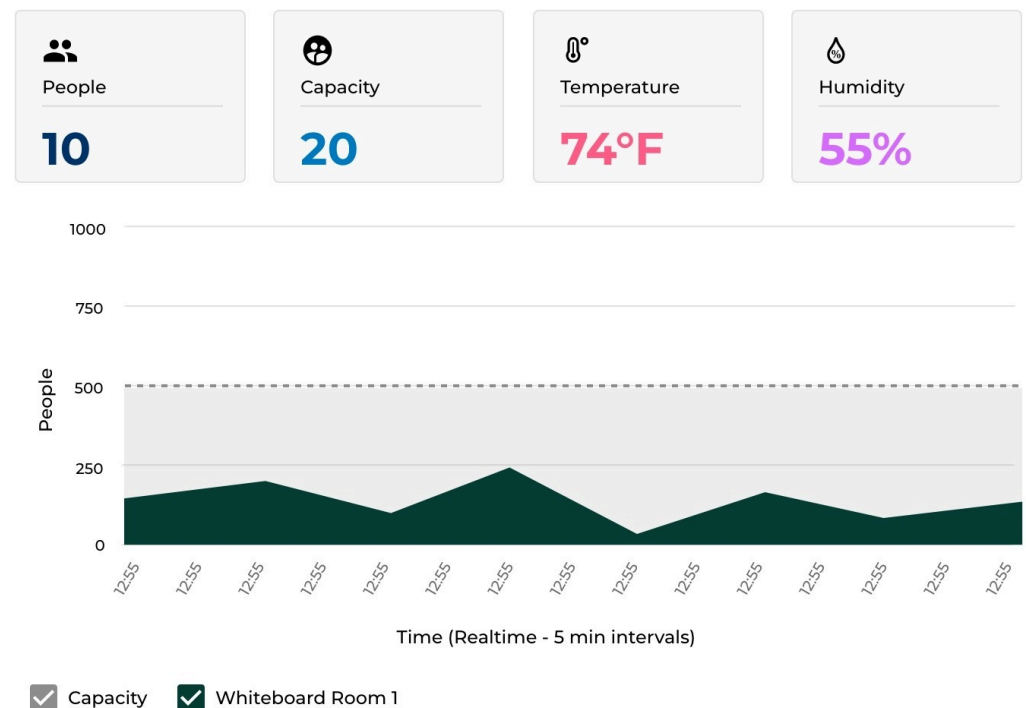


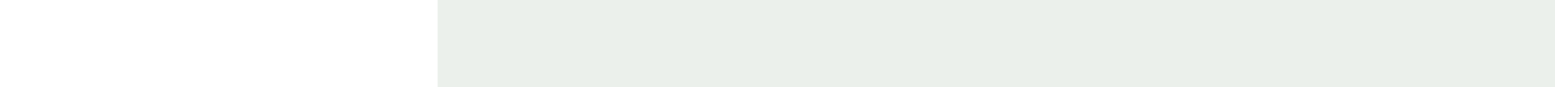
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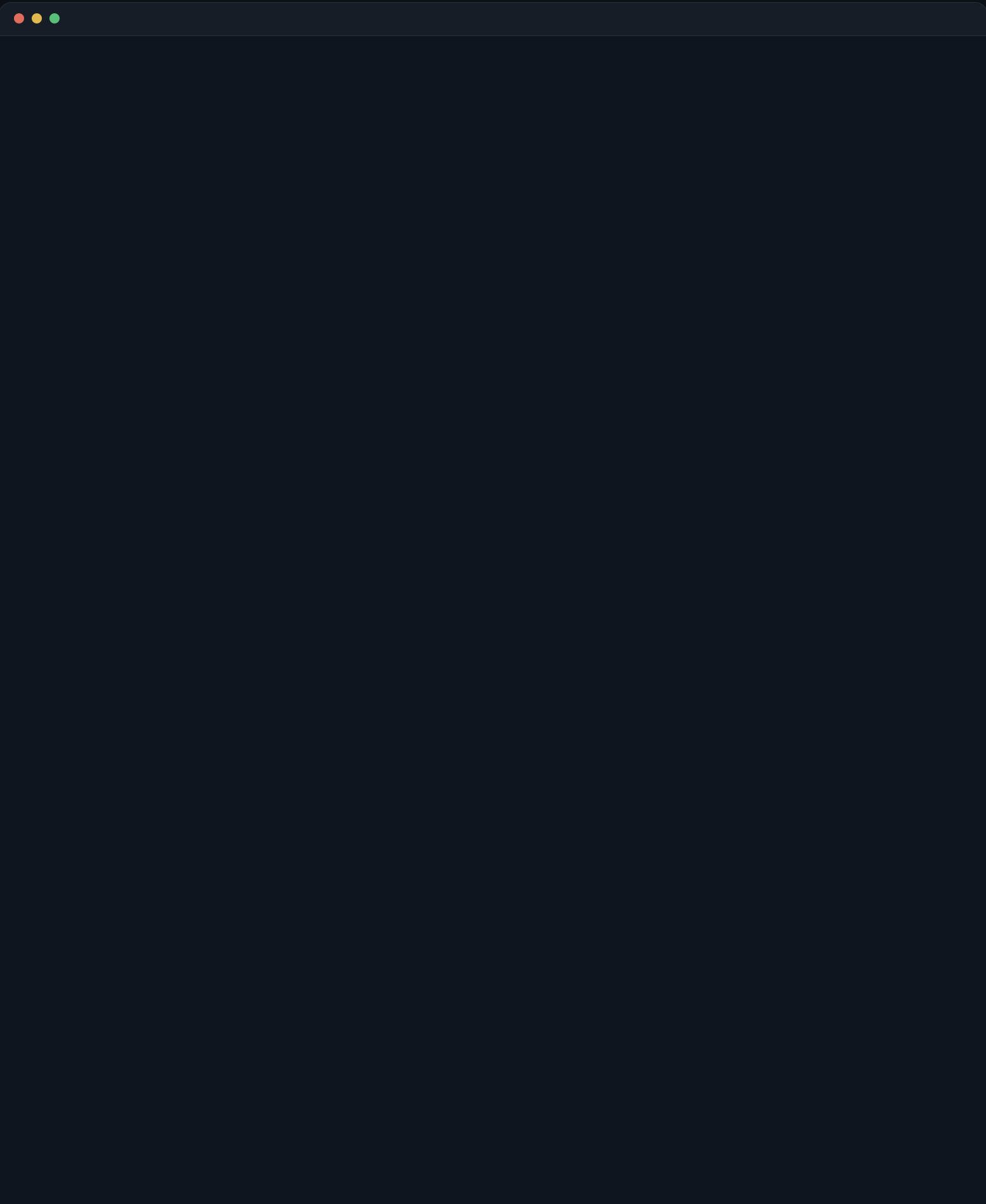
🕒 13:00





CONFERENCE ROOM

Live room status, people, capacity, temperature and humidity, read in spatial context.





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Most Utilization

88%

Floor #5

Most Popular Space

Whiteboard Room

79% Utilization

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(33.33%)

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Average Daily Breakdown

^ SITE

Average utilization for the period, mapped space by space on the floor plan.



Average Daily Breakdown

^ BUILDING



SORT



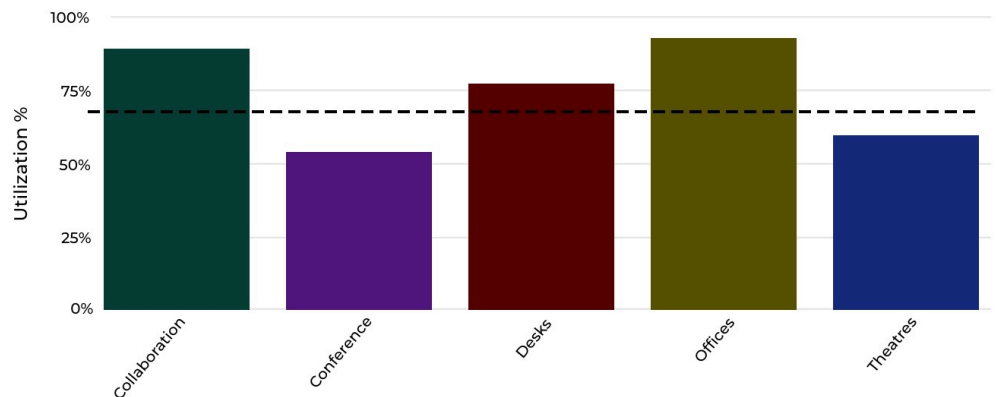
TABLE



Average utilization by space type, measured against the building average.

OVERVIEW

TRENDING



SHOW ALL HIDE ALL

---- Average



Capacity



Collaboration



Conference



Desks



Offices



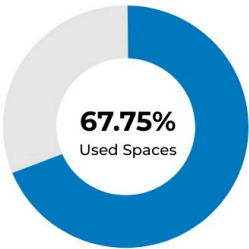
Theatres

The analyst view, floor heatmap, utilization and capacity rolled up on one screen.

Space Utilization

Share of spaces in use vs sitting empty.

32.25%
Empty Spaces
31%
Last Week

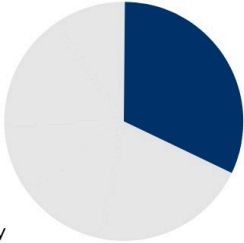


16
Total Spaces

Capacity Utilization

People on site as a share of total capacity.

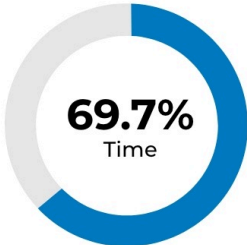
38.29%
Utilization
49.6%
Last Week



69
People
185
Total Capacity

Total Time Space

Space over capacity



Refreshed on April 28, 2020 9:58am

11 • OUTCOMES

0%

Fewer clicks to
root cause



7-9 CLICKS → 1

0%

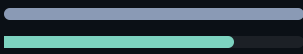
Faster report
generation



5-7 MIN → <30 SEC

0%

Average energy
reduction

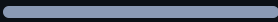
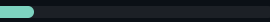
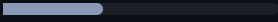
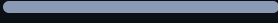
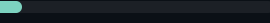
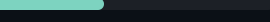
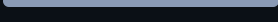
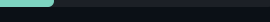


40,000+ fixtures modelled

3 user tiers · 1 system

Web · iOS · Android

USABILITY · BEFORE → AFTER

METRIC	BEFORE	AFTER	IMPACT
Clicks to answer "Which spaces are empty?"	7-9 clicks 	1 click 	-87%
Users drilling past the top-level view	~20% 	~55% 	+175%
Time to generate an executive PDF report	5-7 min 	<30 sec 	-93%
Navigation reset rate (back to top)	~65% 	~25% 	-62%
Floor map → space detail	4-6 clicks 	1 tap 	-80%

EACH ROW'S BARS ARE SCALED TO THAT ROW'S LARGER VALUE · TEAL = AFTER

"CORE Insights changed how our facilities teams operate. They went from monthly reports to a live picture they can act on in seconds."

NEXT CASE STUDY

DALI-2 Lighting System →

rbabiarz@gmail.com[← Back to portfolio](#)

SITE EST

DEPLOYMENTS

LAST DEPLOY

VERSION

Jun 22, 2026

108

Jul 18, 2026

v1.11.113

UPDATED

Jul 19, 2026

BUILT WITH

Anthropic Claude Code · Playwright · Figma + MUI kit · Cursor · Python 3 · React 18 (CDN, no build step) · Git → GitHub Pages

ARCHITECTURE

29 self-contained pages, ~76,000 hand-authored lines, zero npm dependencies, no bundler — every page is a single file that opens anywhere and deploys as pure static hosting. The interactive layer is bespoke vanilla JS: a photometric engine scored against real parking uniformity targets, an AI AutoLayout demo with ranked candidates, playable decision drills, and client-side search over a Python-generated index. WCAG 2.2 AA target with a standing audit; three-tier design tokens (primitive → semantic → component).

AI-PAIRED

Human-led, AI-accelerated: designed and engineered in pair with Anthropic Claude Code, with Playwright browser automation verifying every change — the same receipts-over-vibes workflow the case studies argue for. 11 case studies, several of them about designing AI systems people can trust.

STATS DERIVED FROM GIT HISTORY & THE GITHUB PAGES DEPLOYMENT API, REGENERATED PER RELEASE · VERSION = GENERATION · CASE STUDIES · COMMITS

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